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Studierichting: Informatica
Oefeningenreeks 7B nr. 3.2

$r = uvw - u^2 - v^2 - w^2$ met

$u = y + z,\ v = x + z,\ w = x + y$

$f : (u, v, w) \to uvw - u^2 - v^2 - w^2$

$g : (x, y, z) \to (x + y, x + z, y + z)$

$r = Df(u, v, w) \cdot Dg(x, y, z)$

$Df(u, v, w) = \begin{pmatrix} vw - 2u & uw - 2v & uv - 2w \end{pmatrix}$

$Dg(x, y, z) = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

$\frac{\partial r}{\partial x} = uw - 2v + uv - 2w = (y + z)(x + y) - 2(x + z) + (y + z)(x + z) - 2(x + y)$

$= 2xy + y^2 + 2zx + 2zy - 4x - 2z + z^2 - 2y$

$\frac{\partial r}{\partial y} = vw - 2u + uv - 2w = (x + z)(x + y) - 2(y + z) + (y + z)(x + z) - 2(x + y)$

$= 2yx + x^2 + 2zx + 2zy - 4y - 2z - 2x + z^2$

$\frac{\partial r}{\partial z} = vw - 2u + uw - 2v = (x + z)(x + y) - 2(y + z) + (y + z)(x + z) - 2(x + z)$

$= 2xy + x^2 + 2zx + 2zy - 4z - 2y + y^2 - 2x$

$\Rightarrow \begin{pmatrix} 2xy + y^2 + 2zx + 2zy - 4x - 2z + z^2 - 2y & 2yx + x^2 + 2zx + 2zy - 4y - 2z - 2x + z^2 & 2xy + x^2 + 2zx + 2zy - 4z - 2y + y^2 - 2x \end{pmatrix}$

References